9.8 Taylor and Maclaurin Series

Taylor series - a power series representing a function

Series Representations

A power series is a continuous function with derivatives of all orders. What about the other way round?

Given a function $f(x)$, how do we work out its power series.

Assume that a function $f(x)$ is the sum of a power series

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

$$= a_0 + a_1(x-a) + a_2(x-a)^2 + \ldots$$

with a positive radius of convergence.

Note that $f(a) = a_0$

Suppose $f(x) = \sum a_n (x-a)^n$

By repeatedly differentiating within the interval of convergence, we can work out the terms a_n for each $n \geq 0$. $f(a) = a_0$

$$f'(x) = a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + \dots + n a_n (x-a)^{n-1} + \dots$$

$$\Rightarrow f'(a) = a_1$$

$$f''(x) = 2a_2 + 3.2.a_3(x-a) + \dots + n(n-1)a_n(x-a)^{n-2} + \dots$$

$$\Rightarrow f''(a) = 2a_2$$

$$\vdots$$

$$f^{(n)}(x) = n!\,a_n + R(x), \text{ where } R(x) \text{ is just an infinite polynomial with } x-a \text{ as its factor \& so } R(a) = 0.$$

$$\Rightarrow f^{(n)}(a) = n!\,a_n \Rightarrow a_n = \frac{f^{(n)}(a)}{n!}$$

So if $f(x)$ can be represented as a power series, then

$$f(x) = \boxed{\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n}$$ ← Taylor series generated by $f(x)$ @ $x = a$.

If f has a series representation, then the series must be

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots$$

Taylor and Maclaurin Series

Definition: Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots.$$

The **Maclaurin series generated by** f is [Special case of Taylor series at $x = 0$].

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(a)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(n)}(0)}{n!}x^n + \ldots,$$

the Taylor series generated by f at $x = 0$.

Note: We know that the power series $\sum_{n=0}^{\infty} (-1)^n x^n$ converges to $\frac{1}{1+x}$ in the interval $x \in (-1, 1)$. So the Taylor series generated by $f(x) = \frac{1}{1+x}$ at $x = 0$ is precisely $\sum_{n=0}^{\infty} (-1)^n x^n$.

geometric series with ratio $-x$.

$\Rightarrow \sum \frac{f^{(n)}(2)}{n!}(x-2)^n$

Example: Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$. When does the series converge to $\frac{1}{x}$? We need to find $f^{(n)}(2)$ $f(2) = \frac{1}{2}$

$f'(x) = -\frac{1}{x^2}$, $f''(x) = \frac{2}{x^3}$, $f'''(x) = -\frac{3 \cdot 2}{x^4}$, ...

$f^{(4)}(x) = \frac{4 \cdot 3 \cdot 2}{x^5}$

$f^{(n)}(x) = (-1)^n \frac{n!}{x^{n+1}}$

$\Rightarrow f^{(n)}(2) = (-1)^n \frac{n!}{2^{n+1}}$

Taylor series: $\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!}(x-2)^n = \sum_{n=0}^{\infty} (-1)^n \frac{(x-2)^n}{2^{n+1}} = \frac{1}{2} - \frac{1}{4}(x-2) + \frac{1}{8}(x-2)^2 - \cdots$

This is a geometric series with first term $\frac{1}{2}$ & ratio $-\frac{1}{2}(x-2)$.

So it converges to $\frac{1}{x}$ when $\left|-\frac{1}{2}(x-2)\right| < 1$, so when $0 < x < 4$.

Taylor Polynomials

Taylor polynomials - to approximate a function.

Definition: Let f be a function with derivatives of order k for $k = 1, 2, \ldots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N, the **Taylor polynomial of order** n generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f^{(k)}(a)}{k!}(x - a)^k + \ldots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

Example: The first few Taylor polynomials $f(x) = \cos x$ at $x = 0$ approximate the cosine function near $x = 0$.

$f'(x) = -\sin x$ $f''(x) = -\cos x$ $f'''(x) = \sin x$

$f'(0) = 0$ $f''(0) = -1$ $f'''(0) = 0$

$f^{(4)}(0) = 1$

$$P_0(x) = P_1(x) = 1$$

$$P_2(x) = P_3(x) = 1 - \frac{x^2}{2!}$$

$$P_4(x) = P_5(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

Example: Expand $f(x) = e^x$ at $x = 0$.

$\sum \frac{f^{(n)}(0)}{n!} x^n$

$f^{(n)}(x) = e^x$ for any $n \geq 0$.

$\Rightarrow f^{(n)}(0) = 1$

$$\Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$= \sum_{n=0}^{\infty} x^n / n!$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Example: Expand $f(x) = \ln(1 + x)$ at $x = 0$.

$f(0) = 0$

$f'(x) = \frac{1}{1+x}$

$f''(x) = -\frac{1}{(1+x)^2}$

$f^{(3)}(x) = \frac{2}{(1+x)^3}$

$f^{(4)}(x) = -\frac{3 \cdot 2}{(1+x)^4}$

$\vdots$

$f^{(n)}(x) = (-1)^{n+1} \frac{(n-1)!}{(1+x)^n}$

$\Rightarrow f^{(n)}(0) = (-1)^{n+1}(n-1)!$

Maclaurin series: $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$

$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\cdots$

Example: Expand $f(x) = \sin x$ at $x = 0$.

Maclaurin series : $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$.

$f(x) = \sin x$ $\quad f'(x) = \cos x$

$f''(x) = -\sin x$ $\quad f'''(x) = -\cos x$

$f^{(4)}(x) = \sin x$ $\quad f^{(5)}(x) = \cos x$

$\vdots$

$f^{(2n)}(x) = (-1)^n \sin x$ $\quad f^{(2n+1)}(x) = (-1)^n \cos x$

At $x = 0$ we have $f^{(2n)}(0) = 0$ & $f^{(2n+1)}(0) = (-1)^n$

$$f(x) \text{ "=" } \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{f^{(2n+1)}(0)}{(2n+1)!} x^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Exercise :

Compute the Maclaurin series generated by $f(x) = \cos x$.

Ans: $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$

$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

Example - A function f with converging Taylor series, but not equal to f itself.

$$f(x) = 0, \qquad x = 0$$
$$= e^{-1/x^2}, \quad x \neq 0$$

f has derivatives of all order at $x = 0$ and $f^{(n)}(0) = 0$ for all n.

The Taylor series generated by f at $x = 0$ is

$$\sum \frac{f^{(n)}(0)}{n!} x^n = 0$$

The series converges to 0 for all x.

But $f(x) \neq 0$ @ $x \neq 0$.

$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$

Methods to generate power series

Recall that we can substitute a function $f(x)$ for x in a convergent power series. This method can be used to generate Taylor series for complicated functions.

Example: Find the Taylor series of $f(x) = \frac{1}{1+x^3}$ at $x = 0$.

We first find Maclaurin series of $\frac{1}{1+x}$.

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n = 1 - x + x^2 - x^3 + \cdots$$

$$\Rightarrow \frac{1}{1+x^3} = \sum_{n=0}^{\infty} (-1)^n (x^3)^n$$

$$\therefore \text{Maclaurin series of } \frac{1}{1+x^3} \text{ is } \sum_{n=0}^{\infty} (-1)^n x^{3n} = 1 - x^3 + x^6 - x^9 + \cdots$$

Example: $f(x) = \ln(\cos x), x \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Find the Maclaurin series generated by $f(x)$ up to x^6 term.

We know that • $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

• $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$

$$\ln(\cos x) = \ln[1 + (\cos x - 1)]$$

$$= (\cos x - 1) - \frac{(\cos x - 1)^2}{2} + \frac{(\cos x - 1)^3}{3} - \frac{(\cos x - 1)^4}{4} + \cdots$$

$$= \left(-\frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right) - \frac{1}{2}\left(-\frac{x^2}{2} + \frac{x^4}{4!} \cdots\right)^2 + \frac{1}{3}\left(-\frac{x^2}{2} + \cdots\right)^3$$

$$= -\frac{1}{2}x^2 + \left(\frac{1}{4!} - \frac{1}{8}\right)x^4 + \left(-\frac{1}{6!} + \frac{1}{48} - \frac{1}{24}\right)x^6 + \cdots$$

$$= -\frac{1}{2}x^2 - \frac{1}{12}x^4 - \frac{1}{45}x^6 + \cdots$$

Example - Indirect Expansion:

Find the Taylor series of $f(x) = (1 + x)e^x$ at $x = 0$.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$x + \frac{x^2}{1} + \frac{x^3}{2!} + \frac{x^4}{3!} + \dots$$

$$f(x) = e^x + x e^x$$

$$= \sum_{n=0}^{\infty} \frac{x^n}{n!} + x \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} + \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}$$

$$= 1 + \sum_{n=1}^{\infty} \left(\frac{1}{n!} + \frac{1}{(n-1)!} \right) x^n$$

$$= 1 + \sum_{n \geq 1} \frac{n+1}{n!} x^n = \sum_{n=0}^{\infty} \frac{n+1}{n!} x^n$$

Example:

$f(x) = 1 + x + x^2 + \frac{2}{3}x^3 + \frac{1}{2}x^4 + \dots$

Find the Taylor series of $f(x) = \frac{e^x}{\cos x}$ at $x = 0$. up to x^4 term.

$f(x) = \sum_{n=0}^{\infty} a_n x^n$

$e^x = \sum_{n=0}^{\infty} x^n/n!$ $\qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

$f(x) = e^x/\cos x \Rightarrow e^x = f(x)\cos x$

$\Rightarrow \sum_{n\geq 0} \frac{x^n}{n!} = (a_0 + a_1 x + a_2 x^2 + \dots)(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots)$

$\Rightarrow 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots = a_0 + a_1 x + (a_2 - \frac{a_0}{2})x^2 + (a_3 - \frac{a_1}{2})x^3 + (a_4 - \frac{a_2}{2} + \frac{a_0}{4!})x^4 + \dots$

Compare coefs to get

$a_0 = 1$, $a_1 = 1$, $a_2 - \frac{a_0}{2} = \frac{1}{2} \Rightarrow a_2 = 1$, $a_3 - \frac{a_1}{2} = \frac{1}{6} \Rightarrow a_3 = \frac{2}{3}$

$a_4 - \frac{a_2}{2} + \frac{a_0}{4!} = \frac{1}{4!} \Rightarrow a_4 = 1/2$.

Exercise:

Find the Maclaurin series of the following functions.

① $f(x) = \dfrac{x^2}{1+3x}$

② $f(x) = \sqrt{1+x}$.

↳ $f^{(n)}(0) = \dfrac{(-1)^{n-1}(2n-2)!}{2^{2n-1}(n-1)!}$